56

FINAL INSPECTION TIME 0 TO 10 Kg/cm²
PRESSURE BUILD UP TIME.
CP I :→ 5 MIN 58 SEC.
CP II :→ 6 MIN 08 SEC.
SOD - 15/8/25

SABARMATI DIESEL S. LOCO CARE CENTRE, SABARMATI
Mechanical
IA/IB/IC Schedule of 3Phase Electric Locomotive (Inspection)

लोको संख्या Loco No 32456 क्लास Class WAG9 शेड्यूल में लेने का दिनांक Date taken under Sch 03-08-25

	आगमन Incoming / शिड्यूल Schedule		अंतिम निरीक्षण Final Inspection	
	अपर डैक Upper Deck	अंडर ट्रक Under Truck	अपर डैक Upper Deck	अंडर ट्रक Under Truck
Date/ Shift	03.08.25 0/8	04/08/25 8/17	14.08.25 8/17	14/8/25
Name of Technician	Manoj + Radhe	Ashutosh + Ashok	DIPAK + I.	Ketan
Designation/ Token No.	MCF + Kh.	Tech-1st + APP	TECH-I.	MCF
Signature	Manoj	AKsingh	Solanki Dipak	Ketan

Customer Feed Back/Bookings (Repeat if any booked by loco pilot)

Sr. No.	Customer Feed Back/Bookings	Action Taken	Name of TCN / Attended By	Remarks, if any & Sign. of JE/SSE
	N.A			

Inspection Wing Mechanical: Check and record Following

	CPA Loading Time (Pressure 0 to 8.0 kg/cm²)	Safety Valve Blowing	Pantograph-1		Pantograph-2		Compressor Loading Time (Pressure 8 to 9.5 kg/cm²)		CR Safety Valve Blowing	Unloader valve
			Raising	Lowering	Raising	Lowering	Comp-1	Comp-2		
Std. Value		8.5± 0.25 kg/cm²	6 to 10 Second	Less than 15 Second	6 to 10 Second	Less than 15 Second	[illegible]	[illegible]	10.5± 0.35 kg/cm²	Operation time Approx. 12 Sec.
CI		9.2	7 Sec.	9 Sec.	6 Sec	7 Sec	65 Sec	ISOLATED.	11 L	6 Sec
Final		8.7.	8 SEC.	6 SEC	6 SEC.	8 SEC.	54 SEC	56 SEC.	>11	6 SEC

	MR safety Valve pressure		Duplex check valve setting/Brake Pipe Pressure by Auto Brake (A9)		Feed Pipe Pressure		Brake Cylinder Pressure by Direct Brake (SA9)		Horn Sound			
	Blowing	Closing							Std. - Good			
Std. Value	10.5± 0.2 kg/cm²	9.5–9.8 kg/cm²	5.0 ± 0.1 kg/cm²		6 ± 0.1 kg/cm²		3.5 ± 0.1 kg/cm²		Cab-1		Cab-2	
			Cab-1	Cab-2	Cab-1	Cab-2	Cab-1	Cab-2	LP	ALP	LP	ALP
CI	11.19	10.2	5.0	5.0	6.0	6.0	3.5	3.9	"	"	"	"
Final	10.7	9.8	5.0	5.0	6.0	6.0	3.5	3.5	GOOD	GOOD.	GOOD	GOOD

	Computerized Control Brake (CCB) System if applicable		Air Flow	Air Dryer (Parag POLYMER)		ADC	MRC time By both CP 0-10 kg/cm²
Std. Val.	E-70	Alerter and Penalty	Working	Working, Tower to change every two minutes (FTIL, SIL & in every two minutes (Knorr))	Humidity indicator Colour Std-Blue	Working	5.5 Minutes (Max.) For 1750 LPM or (8.5 Minute for each 1450 LPM CP)
	Passed successfully	Working					
CI		W	Working	Working	Blue	working	2 Min 34 Sec.
Final			W	WORKING	BLUE	W	2 MIN 36 SEC

Signature 03/08/25

	CPA Pressure build up time 8.5 kg/cm²	BP drop in 5 Minute	FP drop in 5 Minute	MR drop in 15 Minute	Press BPPB to apply or release parking brake. Check Pressure in Parking Brake gauge		VCB Pressure switch setting
Std. Value	120 ~~60~~ Sec. Max.	Max. drop 0.15 kg/cm²	Max. drop 0.70 kg/cm²	Max. drop 1.0 kg/cm²	6.0± 0.15 kg/cm²		Opens 4.5± 0.15 kg/cm², Closes 5.5± 0.15 kg/cm²
					Cab-1	Cab-2	
GI	82 sec	0.5	0	0.7			
Final	72 SEC.	NO DROP	NO DROP	0.5 DROP.	NA		Dummy

	ACP on 4mm test plate		Auto brake controller Position		BP (kg/cm²) Cab-1	BP (kg/cm²) Cab-2		BC (kg/cm²) Cab-1	BC (kg/cm²) Cab-2
Std Values	Working		Run	5.0 ± 0.10	5.0	5.0	0.0	0.0	0.0
			Initial Application	4.6 ± 0.10	4.6	4.6	0.40 ± 0.10	0.4	0.4
	Cab-1	Cab-2	[illegible]	3.35 ± 0.20	3.5	3.5	2.50 ± 0.10	2.5	2.5
			Emergency	Below 0.30	0.0	0.0	2.50 ± 0.10	2.5	2.5
Gen. Insp	W	W							
Final	WORKING	WORKING.							

Coaction not working during IRC.

	Loco pilot cabin condition		Wiper Operation		Look out glass		Fire Extinguisher	Sand box	Sander Operation		Cleaning of Loco
	Cab-1	Cab-2	Cab-1	Cab-2	Cab-1	Cab-2			Cab-1	Cab-2	
Std. Values	Good	Good	Working	Working	Clear	Clear	Sealed	Full	Working		Cleaned
Gen. Insp / Final	D	D		Working	D	D	Sealed	EMPTY	NW	NW	D

S.N. SN	कार्यवाही कार्य /विवरण का निरीक्षण / (शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्यवाही की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
(A)	Pneumatic System (When Loco is energised check, attend and record)			
1	Check correctness of reading of BP, FP, MR, BC, AF & SS gauges			
2	Check on dummy of BP/FP pipe.	Check		
3	Check the working of PTDC E-70			
4	Check the proper functioning of all the gauges of the driver desk			
	BP: (i) Normal (ii) with 2.5 mm hole BP should not drop below [illegible] kg/cm² in 60 second	4.7 kg/cm²		
	FP: (i) Normal (ii) with 5.5 mm hole FP should not drop below 4.5 kg/cm² in 60 second	5.8 kg/cm²		
	MR: (i) Mainline MR leakage without application of SA-9 Direct Brake (ii) Mainline MR leakage with application of SA-9 Direct Brake	Cheak ok		
5	RS drop) Panto line air leakage 0.70 kg/cm² in 5 minutes.			
6	Check working of Panto selection switch.	check		3/8
7	Position of isolating cock on pneumatic panel (70, 74, 136 in open condition & 47 in close condition)			
8	Drain moisture from MR and inter Cooler.			
9	Check functioning of air dryer for purging cycle time and change over with MR pressure of 8.0 kg/cm².	Cheak		
10	Check Position of the Air dryer cock (13, 14 in service & 137 in isolated)			
11	Check air dryer charging when compressor is in working. (FTIL-120 sec/SIL-180 sec)	check		
12	Check and attend leakage of air dryer, compressor valves, Pneumatic panel, pneumatic pipes & pipelines.			

Signature of QAG JE/SSE

क्र.स. SN	कार्यवाही कार्य /विवरण का निरीक्षण /(शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्यवाही की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
B	**BRAKE CONTROLLER (AUTO AND DIRECT)**			
1.	Check the functioning of driver's direct air brake & automatic train brake.	Due to co-action not wkg		
2	Check the free movement of the brake controller.			
3	Perform repeated application and release of brakes with both auto and direct brake controller from both cabs and ensure that BP & BC pressure build up & drop is proper on the gauge. Ensure physically going to individual main and parking brake cylinder that their operation is smooth and without air leakage			
C	**CARBODY FILTER:** Check condition and clean car body filter and refit.	Checked		
1	Thoroughly clean all roof and side body filters with water jet.	cleaned		
2	Check tightness of Roof joint clamp and Roof top bolt.	checked		Ambitsi 6/8/25
3	Remove Louvers& Fly screen and clean it. Repair if found defective	~~Chec~~ —		
	IC sch.			
4	Remove the tread plate and clean the drainage channels Check and adjust, if necessary, the door-operating rod.	Clean		
H	**LOOKOUT GLASS & SIDE WINDOWS**			
1	Clean any built up dirt and debris from the channel below the cab-sliding window. Clear the sliding window channel drain hole in the door opening.	checked by cab fitting staff		
2	Clean look out glass with detergent/water solution			
I	**WASHER / WIPER**			
1	Check the operation of windscreen wipers using the manual handles. Rectify any fault found.	Check ok		
2	Check the condition of windscreen wiper blades. Replace if required			
4	Check the water level in the reservoir at No.1 and No.-2 end Top up if required. Check wipers for proper operation			
6	Check for damage of air and water pipes	Check ok		
7	Examine all components for wear, looseness scoring and cracks, renew any defective components.		3/8	
8	Check seat-mounting points for integrity and ensure fixing of fully tightened	Check ok		
J	Key interlocking: Check all keys are present & the complete system is operational			
	(IC Sch)			
9	Check the wiper and parallel arm assembly for wear and damage. Replace the arm or any part if there is excessive wear or damage			
10	Clean the washer nozzle jet. **(IC Sch)**			
K	**Fire Extinguisher**			
1	Ensure availability of fire extinguishers and ensure that it is not expired.	Check ok		
2	Check the condition of CO2 gas cylinder, its regulator and pressure gauge in both cabs. Check the CO2 gas pressures. Refill if necessary	Check ok		
3	Clean pipeline by compressed air and check that there should not be any leakage.		3/8	
4	CO2 Gas pressure and date: Cab-1 70 Cab-2 70	Check ok		
5	Clean fire detection unit pipeline air intake holes with soft non-metallic brush & vacuum cleaner.			

EXP. Date
Cab 1 => 06/12/29.
Cab 2 => 06/12/29

Signature of SSE/JE/SSE

क्र.सं. SN	कार्यवाही कार्य /विवरण का निरीक्षण / (शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्यवाही की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
L	OIL LEAKAGE IN MACHINE ROOM / CLEANING OF MACHINE ROOM			
1	Visually inspect the machine room for any signs oil leakage fromtraction converter, transformer, D.C. Link, and series resonant circuit capacitors and other electrical equipment and carefullyclean the same. (Caution : Electric supply room)	Check o/c		
2	Clean the complete machine room with the air pressurizedsupply and vacuum cleaner to remove all the dust particles.			
M	SAND GEAR SYSTEM			
	a) Check Sand box, cover gasket, Ejector and replace gasket if necessary.	Check o/c	378	
	b) Check all sander pipe for alignment and check fasteners also			
	c) Check and ensure tightness of Sand box foundation bolt			
	d) Ensure sander pipe height 60mm above rail level			
	e) Refill sand in all sand boxes.			
N	OIL COOLING UNIT CASING WITH RADIATOR			
1	Remove all dust, dirt and debris from the radiator chamber viathe machine room access cover. (By using vacuum cleaner)			
2	Visually check the conservator tank along with its gauge glassfor any sign of crack / damage / oil leakage	Checked		
3	Measure air delivery before and after cleaning of radiator.	Checked		
4	Measure vibrations of oil cooling casing using vibration analyzerand take appropriate action	Checked		
5	Visually check the casing and support arrangement for oilpumps on the casing for any sign of crack of the damage	Checked		
	Visually check the oil cooler radiator for any oil leakage from VPTFP and VPSR and externaldamage from top to bottom	Checked		
	IC Sch			
7	a) Clean radiator with hot water jet and check that the radiator is not blocked.	Checked		
8	b) Tighten the front foundation bolts of casing mounted on to the radiator and check intactness of jali and Z-clamp welding crack.	Checked		

Check and put (√) mark on the check box (CB) against each of the following items if the same has been checked. Write defects noticed and action taken in the Non-Schedule work table.

UPPER DECK [Inspect, Attend and Record]

SN	Description	CB	Remarks
1.	Loose, missing and broken components/parts	✓	
2.	Clamping and condition of its welding/fastener.	✓	
3.	Condition and working of all pressure gauges	✓	
4.	Working of all brake valves.	✓	
5.	Leakage of air (MR)	✓	
6.	Gauges, securing of loco pilot cabin	✓	
7.	Operation of Wipers.	✓	
8.	Co-Action working of A9 Auto brake.	✗	Not wkg
9.	Fire Extinguishers (prescribed Nos.)	✓	
10.	Check tightness of compressor foundation bolt & abnormal sound.	✓	
11.	check slackness of compressor fan and no vibration in comp.	✓	
12.	Check tightness of air dryer foundation bolt.	✓	
13.	Ensure Air dryer indicator blue colour	✓	
14.	Ensure clamping of all pipe to avoid vibration and rubbing.	✓	
15.	Ensure fitment of all split pins	✓	
16.	Check tightness of all bolts of Loco Roor	✓	
	Check window of both cab, shutter window door & corridor doors	✓	

Signature of QAG JE/SSE

03/08/20

Check and put (√) mark on the check box (CB) against each of the following items if the same has been checked. Write defects noticed and action taken in the Non-Schedule work table.

UNDER TRUCK [Inspect, Attend and Record]

SN	Description	CB	Remarks
1.	Loose, missing and broken components/parts.	✓	
2.	Clamping and condition of its welding/fastener.	✓	
3.	Air leakage in bogie.	✓	
4.	Overheated parts / Abnormal wear in parts.	✓	
5.	Check and Ensure Brake block and Wheel clearance >10mm	✓	

(Remarks if any & sign): 1. Coaction not working during IRC
2. SR-01 bottom oily.
3. CP2 cut out at 8.5 kg/cm² pressure only.

Signature and Name of Technician for Inspection Incoming with date & shift: Manoj + Radhamohan

Signature and Name of JE/SSE for Inspection Incoming with date & shift: Rajesh Chand 03/08/25, 08 shift

Signature and Name of Technician for Inspection Final with date & shift: Dipak | DIPAK / VISHAL | 14.8.25 / 8/17 SHIFT.

Signature and Name of JE/SSE for Inspection Final with date & shift:

अन्य किसी भी जानकारी Any Other Information:

Signature of QAG SSE Incharge

SCHEDULE PROGRESS CHART

SN	Date	Shift	Work Details	Name of JE/SSE	Sign of JE/SSE	Remarks
1	03/08/24	O/B	IRC Clout	Rajesh	[illegible]	
2	04/08/25	16/2A	Com. sch. done			
3	14.8.25	8/17	FINAL-INSPECTION			
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						
17						
18						
19						
20						
21						
22						
23						

SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI
IA/IB/IC SCHEDULE OF 3 PHASE ELECTRIC LOCOMOTIVE [UNDERFRAME]

लोको संख्या Loco No	32456	क्लास Class	IC	शिड्यूल में लेने का दिनांक Date taken under Sch	7/8/24

UNDERFRAME IA/IB/IC SCHEDULE & RECORDS

A-Check and record **Hoot Wear, Flange Wear & Tread Wear** (Record Wheel Diameter in Schedule or After TT)

Wheel No	Wheel Tread dia	Root Wear	Flange Wear	Trean Wear	Wheel No	Wheel Tread dia	Root Wear	Flange Wear	Tread Wear
1	1062.1	4.0	1.5	3.0	7	1063.5	3.5	1.0	3.5
2	1063.0	4.0	2.0	3.0	8	1062.1	3.0	1.0	2.5
3	1062.8	3.0	0.5	3.5	9	1062.8	1.5	0.0	3.0
4	1063.1	2.5	0.5	2.5	10	1063.9	2.0	0.5	2.0
5	1061.9	4.0	1.5	3.5	11	1063.9	3.5	1.0	3.0
6	1062.1	4.0	2.0	3.0	12	1065.8	4.0	1.5	3.0

B- Check and Record Vertical & lateral clearance (Primary & Secondary) as per TC/0082 (REV-1) dated 17/09/2015

Standard Value	Primary vertical clearance (Cab 1 & 2 Bogie) Actual Value Axle box & Bogie frame											
	1	2	3	4	5	6	7	8	9	10	11	12
27 to 35 mm	33.50	36.35 / 29.30	35.00	31.25	35.90 / 28.85	34.70	31.95	36.65 / 32.35	33.40	33.40	34.95	35.00

6/8/24

Standard Value	Primary lateral clearance (Cab-1&2 Bogie) Actual Value Axle box & Bogie frame											
	1	2	3	4	5	6	7	8	9	10	11	12
15 to 22 mm	16.55	15.90	16.95	15.00	16.00	15.00	15.05	16.35	16.50	15.00	16.90	15.00

Standard Value	Secondary Vertical clearance (Cab-1 Bogie) Actual Value Bogie frame & under frame			
	CAB-1 LP Side	CAB-1 ALP Side	CAB-2 LP Side	CAB-2 ALP Side
32 to 60 mm	52.35	41.50	50.90	41.85

Standard Value	Secondary lateral clearance (Cab-1 bogie) Actual Value Bogie frame & under frame			
	CAB-1 LP Side	CAB-1 ALP Side	CAB-2 LP Side	CAB-2 ALP Side
45 to 55 mm	41.50 +	48.85	37.65 +	53.90

$\frac{90.35}{2} = 45.17$ $\frac{90.95}{2} = 45.47$

C- Check and record wheel gauge below Parameters: (IC Schedule) WHEEL GAUGE [Permissible limit 1596±0.5 mm (1596±0.5 mm for RB/New loco)]

Axle No.	A	B	C	Average	Remarks
1					
2					
3		NA			
4					
5					
6					

Signature and Name of Technician

Signature and Name of SSE

3

Check and record below Parameters in Schedule on IT ... e

Axle No.	Standard (mm)		Cab-1		Cab-2	
Location	Max	Min	R1	L1	R2	L2
Buffer Length	635	605	635	635	635	640
Buffer Height	1105	1030	1105	1108/1100	1110/1105	1103
Rail guard height	113	105	109	117	100	106
CBC Height	1105	1030	1077	-	1093	-

		Cab-1	Cab-2
STRIKER WEAR PLATE	New 8 mm, Condemn 2 mm	6	7
SHANK WEAR PLATE	New 6 mm, Condemn 1 mm	7	6

E. Measure D SHAKLE and write Coupling No.

CAB-1 OK CAB-2 OK

Check and Record Gear case Oil Level

TM	OIL FOUND	OIL TOP UP	TM	OIL FOUND	OIL TOP UP
1	.00	7	4	6	6.0
2	0.0	6.0	5	6	6.0
3	0.0	7	6	60	6.0

[illegible] 1. 3. 6. [illegible]

	Hand Brake Operation	CB Coupler Operation & Condition		Screw Coupling Operation & Condition		CBC Lock Pin		BP/FP Pipe	
Std. Value	Working	Normal		Normal		Intact		Intact	
		Cab-1	Cab-2	Cab-1	Cab-2	Cab-1	Cab-2	Cab-1	Cab-2
Gen. Insp.	OK	OK	OK	OK	OK	OK	OK	OK	OK
Final Insp.									

	Brake Piston Travel						Brake Shoe release	Standard 3/9/10 mm
Class of Loco	Standard Value	Observed (mm)						
WAG7/WAP7	101 to 117 mm	Location	1	2	3	4		
WAG9		Loco No.	110	111	111	110		

Sr. No.	Detail of Work / Inspection	Action Taken	Sign/Remark
F	DRAFT GEAR AND COUPLERS		
	1. Liberally apply general-purpose grease to the face of the center buffer coupler at the end of the Locomotive	check ok	
	2. Check the intactness of the CBC bottom support plate bolts	check ok	
	3. Check intactness and functioning of CBC operating handle	check ok	AKSingh 04/08/25
	4. Check visually the striker casting of CBC assembly	check ok	
	5. Check for working of CBC lock and smooth locking arrangement working of TC	check ok	
	6. Check washer lock at knuckle pin bottom	check ok	
	7. Check gap between Striker and Draft gear for draft gear slackness	check ok	
	IC Sch.		
	8. Check the knuckle against crack by DPT	—	
	9. Check knuckle by gauge no. ... (... gauge)	—	
	10. Check knuckle nose wear by gauge no. ... gauge	—	
G	BUFFERS		
	1. Check buffers for any crack visually of the ... and intactness of foundation bolts, check nuts and split pins, if not ...	check ok	AKSingh

NOT [illegible] (5) [illegible]

Signature and Name of JE/SSE

[illegible] 6/8/25 [illegible] SE/17

Sr. No.	Detail of Work/ Inspection	Action Taken	Sign/Remark
	2. Liberally apply general purpose grease to the ... of the Locomotive. Also smear grease on the buffer ...	check ok	Akash
	IC Sch.		
	3. Conduct RDPT of Buffers to detect crack flaws	Checked	Akash (noted)
	4. Check and measure the buffer height and buffer ...		
H	**SUSPENSION**		
	1. Examine the primary and secondary springs for ... broken end visually with torch light and tapping ... necessary	check ok	Akash 04/08/25
	2. Examine different types of dampers for ... fasteners and end fittings	check ok	
	3. Check the condition of various Spiters block ... the primary and secondary suspension arrangement and of various equipment ... under frame like axle guide link, traction motor ... support arm for their healthiness (it should ... necessary	check ok	
	IC Sch.		
	4. Proper torque of all fasteners related to ... mounting of equipment in under frame	—	
	5. RDPT of all TM nose and its support arm		
I	**TRACTION BAR**		
	1. Check the well being and presence of the safety ...	check ok	Akash
	2. Check the push/pull rod pivot heads for damage	check ok	
	3. Examine the traction link assembly for signs ... should be renewed	check ok	
	4. Check the fastener at the traction ends and the ... for security. If a fastener is loose, it must be ... specified torque. Ensure intactness of the locking ... MP241	check ok	
	5. RDPT of traction bar at end (flange) portion & ...	check ok	Akash
	6. Housing flange		
	7. Examine limit chain and links for wear, corrosion ...	check ok	
	8. Inspect the safety slings, pins and R clips for wear ...	check ok	
	9. Check Aclathan/Vulkollon ring against any crack ...	check ok	
	10. Check gap in housing flange and traction link flange		
	11. Torque modified bolt M20 X65 by 260Nm (SMI 241)		
	IC Sch.	—	
	12. RDPT of Pivot post for any crack		
J	**BOGIE FRAME AND BRAKE GEAR**		
	BRAKE BLOCK		
	1. Inspect the brake blocks for wear. If it is below the minimum limit, replace the complete locomotive set brake block	check ok	Akash 04/08/25
	2. Check the brake block key security	check ok	
	3. Check the brake rigging for loose, damaged or worn out parts. Rectify as required.	check ok	
	4. Examine the bogie frame, transom beam & damper base for signs of crack/damage	check ok	
	5. Check fitment of Palm Pull rod J Bracket, hanger lever and TM safety rope wire	check ok	
	6. Inspect the rubber bumps for damage deterioration. Replace if required	check ok	
	7. Examine all bogies mounted equipment (including earthing straps) for signs of damage and security of fastening	check ok	
	8. Inspect the brake rigging rotation points and component contact surfaces. Rectify any faults found.	check ok	
	9. Check the earthing shunts provided on bogie and body. Check intactness of axle box cover bolts and sealing	check ok	

Signature and ... of ...

Sr. No.	Detail of Work/ inspection	Action Taken	Sign/Remark
	10. Inspect brake gear for any worn out loose, damaged, hanging or dragging parts. Confirm intactness.	checked ok	
	11. Test for correct operation of the brake cylinder actuator unit.		
	12. Check the proper tightness of parking brake & service brake pneumatic pipes.	N.A	
	13. Check the external hitting marks & physical damage on the brake cylinder actuator unit.	checked ok	
	14. Check the condition of parking brake cylinder bellows if damaged, replace them.	N.A	
	15. On parking brake cylinder pulling spindle & ring should be intact.	N.A	
	16. Check proper functioning of parking brake & service brake cylinder.	checked ok	
	17. Check the operation (manual releasing) of the parking brakes	N.A	
	18. Check the bogie isolating cock, should be in proper position.	checked ok	
	19. Check the wear of scrubber blocks, replace if required (WAP5only)	N.A	
	20. Maintenance of Traction Motor support plate and bogie nose to Prevent crack/breakage of Traction Motor support plate (Holder for Traction motor suspension) should be done as per RDSO's SMI no. RDSO/2011/EL/SMI/0269 (Rev. '0') dtd, 18.05.11	checked ok	
	IC Sch.		
	21. Inspect the bogie body safety pin eyehole at the bogie frame for wear or cracks.	checked ok	
	22. Maintain the vertical & lateral clearances (Primary & secondary) as per TC-082	checked ok	
ii	WHEEL SET AND AXLE BOX		
	1. Examine all the wheels visually for following defects		
	a) Cracks: if any, requires replacement.	checked ok	
	b) Flats: Length 40mm or more requires re profiling.	checked ok	
	c) Cavities: Isolated cavities of length 15mm or more or multiple cavities of length 10mm or more and less than 50mm apart require re-profile/replacement.	checked ok	
	d) Metal builds up Height 1mm, or above and length 50mm or more requires re-profiling. For length less than 50mm., remove the build up with hand tools.	checked ok	04/08/25
	e) Scoring and Grooving: Circumferentially around the wheel tread due to action of brake blocks require re-profile/replacement.	checked ok	
	2. Grease Axle box after 300000 KM Engine Run	N.A	
	3. Check the temperature of axle boxes. The axle boxes which are running warm or reported to have higher temperature in comparison with others shall be attended on priority by physically opening of axle box cover and outer angle ring. The suitable maintenance shall be done either in-situ or by dismantling the axle box.		
	IC Sch.		
	4. Measure and record the wheel diameters and tread profile. Refer RDSO's Tech Circular no. ELRS/TC/0083 (Rev. "0").	checked ok	
	5. Disconnect the electrical cable from the earth return unit. Remove the earth return unit from the axle. Examine the bearings and rear sealing position for signs of grease leakage or deterioration.	checked ok	
	6. Check the axle for any cracks with the help of ultrasonic flaw detector. At the same time visual inspection should be done. (IC Sch Every 06 month)	checked ok	
iii	GEAR CASE (WAP7/WAG9/WAG9H)		
	1. Externally clean the gear case & its gauge glass. Inspect for any signs of oil leakage and any external hits/ damage. Verify integrity of gauge glass protective cover and clean oil sight glass for visibility.		
	2. Check the oil level in the gauge glass and top up if required. Ensure intactness of the drain plug along with its sealing. Check its tightness.		
	3. Check for any breakage of bolt and replace		
	4. Check for intactness of all fasteners of gear case	checked ok	
	5. Tighten all gear case fasteners at prescribed torque	checked ok	

Signature and Name of JE/SSE

Sr. No.	Detail of Work/ Inspection	Action Taken	Sign/Remark
6.	Check the gear case oil visually and replace if found dirty or dust full level in the gear case and fill to the requirement	Checked	
7	Check Gear case oil for metal contamination	Checked	
8.	Visually examine for fractures, the tyre profile for wear, flat, and skidding marks		
9.	Check wheel root, flange and Tread wear & record Change all unserviceable Brake Blocks	Checked R.F.T	
10.	Check condition and operation of Hand Brakes. Lubricate & adjust as necessary.	checked	
11.	Adjust Brakes and lubricate union	checked	
12.	Lubricate Brake Gearing, Slack Adjusters, Buffers, CB Coupler Assembly	checked	
13.	Split the cotter pin, if new provided		
	IC Sch.		
14.	a) Axle box old grease to remove and New grease to fill, old grease sample to be sent Lab for metal content check		
	b) Lubricate suspension roller tube bearing.	checked	
	c) Check Transition screw coupling long and short shackle bar thickness by Go-No Go gauge (39.5 mm snap 1/25 mm snap 8)	checked	
	d) Carry out DFT of Buffer Foundation and DPT/MPT of CBC Knuckle clevis &Yoke. Check size of Clevis, Knuckle, coupling by Go-No Go Gauge if specified. As per SMI-213/MIG-76		

Check and put (✓) mark on the check box (CB) against each of the following items if the same has been checked. Write defects noticed and action takes in the Non-Schedule work table.

UNDER TRUCK [Inspect, Attend and Record]

SN	Description	CB No.	Remarks
1.	Measure and record buffer & rail guard height.		
2.	If skidded, fill Wheel Sledding Complain Report Farm, FRMB/22 or if Biased wheel wear, fill up Check last for Based Wheel Wear Cause Invest, FILM/24)	✓	
3.	Proper Fitment of Brake Block and as Key	✓	
4.	Rubbing of brake block with wheel in brake release position	✓	
5.	Movement of brake pull rod in brake applied position	✓	
6.	Brake Shoe and Brake hanger Pin/Bolts		
7.	Brake piston movement.	✓	
8.	Support arrangement for brake cross tie bars	✓	
9.	CBC Drooping.	✓	
10.	Operation of both Transition screw couplings	✓	
11.	Operation of CBC, Operating Rod and Coupler lock Pin	✓	
12.	Operation of CBC Knuckles	✓	
13.	CBC retaining plate and support plate bolts	✓	
14.	Side buffers and their mounting bolts.	✓	
15.	Bolts of Cattle guards and back supports.	✓	
16.	Fitment of Rail Guard	✓	
17.	Gear case mounting bolts and 'C' clamp bolts	✓	
18.	Suspension bearing cap bolts and it's sealing.	✓	
19.	Crack in TM Suspension taper roller bearing housing		

Signature and Name of JE/SSE

After TT Reading

32456

Wheel No	Wheel Tread dia	Root Wear	Flange Wear	Tread Wear	Wheel No	Wheel Tread 1062.5	Root Wear	Flange Wear	Tread Wear
1	1060.0	1.5	.5	.5	7	1062.5	2	1	.5
2	1059.3	.5	.5	1	8	1062.0	.5	.5	1
3	1059.7	1	.5	1	9	1062.9	1.5	.5	1
4	1059.0	1.5	0	.5	10	1063.6	1.5	.5	.5
5	1058.9	1	.5	1	11	1063.6	1	.5	.5
6	1058.1	1	.5	.5	12	1064.0	1	0	.5

Axle No.	Cab 1		Cab-2	
Location	R1	L1	R2	L2
Buffer Length	635	635	635	635
Buffer Height	1100	1100	1100	1100
Rail guard height	107	114	105	105
CBC Height	1070		1085	

[illegible]
14/8/25

Signature and Name of Technician

Signature and Name of JE/SSE

SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI
IA/IB/IC SCHEDULE OF 3PHASE ELECTRIC LOCOMOTIVE
[COMPRESSOR &PNEUMATIC SYSTEM]

लोको संख्या Loco No 32456 क्लास Class WAG-9 शिड्यूल में लेने का दिनांक Date taken under Sch 3-8-25

क्र.सं SN	कार्यवाही कार्य /विवरण का निरीक्षण / (शिड्यूल) Detail of Work/ Inspection (schedule)	कार्यवाही की Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
A	**मेन कंप्रेसर और बेबी कंप्रेसर Main Compressor and Baby Compressor**			
1	Check the pressure build up timing of each compressor independently in pre-inspection and take needful action during inspection. [From 0 to 10 kg/cm^2 i) With 1750LPM Comp-7 Minutes max. ii) With 1450LPM Comp- 8.5 Minutes max.]	CP1 → 6min 10 sec CP2 → getting off @ 08.50kg/cm2 not reaching 10kg/cm2.		
2	Feel the vibrations of the compressor in pre-inspection and take needful action during inspection.			
3	Check for any external damage / abnormal sound / proper functioning of the compressor.			
4	Check the complete compressor support brackets for their healthiness.			
5	Visually examine the compressor mounting resilient blocks and take needful action.			
6	Visually examine the various interconnecting pipelines and its support clamps of the compressors for their healthiness / air leakage			
7	Cleaning of input air filter and the primary oil filter.			
8	Clean the breather assembly			
9	Check and Drain the inter cooler / after cooler assembly and also observe its discharge and take needful action.			
10	Ensure healthiness / intactness of various protection guards of compressor.			
11	Check the condition of the coupling between compressor & motor			
12	Visually check the compressor fan blade for their healthiness			
13	Check the oil level in the compressor and top up as per requirement			
14	Carry out the inspection of each type of compressor in line with recommendation of respective manufacturer.			
15	Check compressor foundation base for crack (For under slung compressor)			
B	**Aux. Compressor (CPA)**	cleaned		
1	Clean the CPA externally. Clean Suction Air filter.			
2	Check oil level and condition of oil and top up if required with the oil specified (Servo Press 150).			
3	Check tightness of all nut bolts, foundation bolts.			
4	Measure and record time & pressure of the auxiliary compressor			
C	**Sch IC**: i) CP all discharge valves to remove.			
	ii) CP all discharge valves refit by overhauled			
	iii) CP oil to be change.			
	iv) Baby compressor/CPA oil to be change			
	CPA			
	CP-1			
	CP-2			
	v) Baby compressor/CPA safety valve (8.5 kg/cm^2) to be replace by overhauled			
D	**AIR SUPPLY AND PNEUMATIC SYSTEM**			
1	NRV and unloader valves fail due to ingress of Carbon which extracted from CP. So, one cycle dismantling, checking and cleaning of NRV and unloader to be ensured in all type of 3-phase locomotives.			
E	**AIR DRYER**			
1	Inspect the air dryer for any damage and its mounting for any damage.			

क.स. SN	कार्यवाही कार्य /विवरण का निरीक्षण / (शिड्यूल) Detail of Work/ Inspection (schedule)	कार्यवाही की Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
	crack in mounting plate, intactness of fasteners. Check for any hit mark on main reservoir and drain cock. Check for proper position of the air dryer angle cocks.			
2	Check the integrity of the welding of the hanging support.			
3	Check humidity indicators for blue color. White indications mean the air dryer is not operating effectively. Desiccant shall be changed, if needed.			
4	Replace desiccant if found to be contaminated with either water or oil.			
5	Testing of air dryer for proper functioning to be done as perTechnical Circular no. **RDSO/2011/EL/TC/0108, Rev.'0' dtd. 7.3.11** for testing of air dryers and its efficacy on electriclocomotives.			
F	**E-70 BRAKE SYSTEM (PNEUMATIC PANEL)**			
1	Visually inspect the pneumatic panel noting any sign of airleakages and rectify accordingly.			
2	Clean the pneumatic panel externally.			
3	Check for correct position of the cut out/isolating cock.			
4	Check all the cable connections for looseness or deficiency orcleats. If any inspection section is to be informed (IC Sch)			
5	Ensure provisioning and proper functioning of RS gauge, Switch, lamp & other gauges on pneumatic panel			
6	Check the sealing of connectors, flexible pipe, regulator, air tank etc.			
G	**BRAKE ISOLATING COCKS:** Check for correct operations of BP & FP angle cocks. Check the condition of BP/FP pipe, cock & isolating handle for any hit markor crack. Check correct condition of Bogie Isolating cock.			
H	**BRAKE HOSES:** Examine the brake hoses at the ends of the locomotive to make sure that they are not chafed or damaged. Check couplingcocks for correct operation.			
I	**AIR RESERVOIR**			
1	Blow down and drain the reservoir until exhaust of complete water and oil. Check air drain cocks are closed on completion.			
2	Check for any hit mark on main reservoir & drain cock			
3	Ensure intactness / tightness of main reservoir foundation bolts.			
4	Check the rubber gasket at the inspection cover of air reservoir for any damage/leakage.			
5	Check the pressure regulator operates correctly			
J	**THROTTLE VALVE** (IC Sch): Overhaul the throttle valve and change the rubber parts.			
K	**IN LINE FILTER** (IC Sch): Remove the filter element and clean thoroughly.			
L	**HORN & HORN SWITCH**			
1	Check condition of horn handle for any damage.Check for proper operation of horn.			
2	Check the condition of horn boots and replace if required			
3	Remove the horn & horn switch and Overhaul. Replace thediaphragm of horn. (IC Sch)			
4	Check all angle cocks for proper operation			
5	Ensure proper colour code of all the angle cocks.			
6	Check the tightness of the brake electronics control cards holding screws.			
7	Check all the pipes for leakage or damage. Check all the clamps for any damage and looseness.			
M	**वायवीय प्रणाली PNEUMATIC SYSTEM**			
1	MR Safety valve, ADC, Panto throttle valve (Only for conventional Pantograph) replace by Overhaul in IC Sch.			

03/08/25

Signature and Name of JE/SSE

SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI
CHECK SHEET FOR 3PHASE HR PANTOGRAPH ASSEMBLY (IA/IB/IC)

लोको संख्या Loco No 42456 क्लास Class WS-5 शिड्यूल में लेने का दिनांक Date taken under Sch 6/8/25

SCHEDULE: .... PANTOGRAPH SR.NO. /Make//Type: .... LOCATION: ....

SN	Detail of Work/ Inspection (schedule)	Standard value	Action Taken PT-1	Action Taken PT-2	Remarks
1.	Ensure that wearing strips are without sharp edges and no gap between end strip and wearing strip.	Smooth and no gap	Smooth and No gap	Smooth and No gap	
2.	Ensure availability of bolts, nuts, lock nuts, spring, washer, split pin etc.	All intact	All intact	All intact	
3.	Check the rubber air pipe for any leakage	No leakage	No Leakage	No leakage	
4.	Check the rubber air bellow for any leakage/damage/rubbing	No leakage	No Leakage		
5.	Check lower frame assembly for any crack and play	No crack	No Crack		
6.	Check upper frame assembly for any crack and breakage	No crack	No Crack		
7	Check coupling rod assembly for crack and its fitting intactness.	No crack & intact	No Crack intact		
8.	Check diagonal rod for any crack both end side. (by DPT)	No crack	NO Crack		
9.	Check plain bearing eye bush for freeness.	free	free	free	3010142 06/08/25
10.	Check ball bearing friction of Base frame, Upper frame and Coupling rod.	ok	ok	ok	
11.	Check Cam wear on top of cable rope guide surface.	OK	ok	ok	
12.	Check condition of all shunts (not more than 5% strand broken)	OK	ok	ok	
13.	Check pantograph pan for any external hitting marks	No hitting marks	No hitting Marks	No hitting Marks	
14.	Verify all welded joint for cracks.	OK	ok	OK	
15.	Check the control box for any air leakage	No air leakage	No air Leakage		
16.	Check the condition and operation of hydraulic damper.	Normal	Normal	Normal	
17.	Check Parallel guide bar for free function of Panto Pan Head	working	Working	Working	
18.	Ensure that the thickness of wearing strips above the condemning size	New – 39 mm Condemn – 26 mm	39.95 – 39.67	32.74 30.87	Cab-1 New Provided
19.	Check the static contact force of pantograph at 6.4 kg/cm^2 MR pressure	70 N / 7Kg Load Test	ok	ok	
20.	Check the raising and lowering time of pantograph	6 to 15 second		R- 13sec L- 15sec	
21.	Check the horizontality of panto pan with sprit level.	Level OK	ok	ok	
22.	Check the function of auto drop device (ADD)	Working	+	—	

SN	Detail of Work/ Inspection (schedule)	Standard value	Action Taken		Remarks
			PT-1	PT-2	
23.	Check the function of over reach detector (ORD)	Working	W —	—	
24.	Check air filter for air leakage or not draining.	No leakage	No Leakge	No Leakage	
25.	Check the all pneumatic valve (3/2 valve, throttle valve and safety valve) for any leakage and proper function.	No leakage	No Leakge	No Leakage	
26.	Check pressure regulator for any leakage and proper function.	No leakage	No Leakge	No Leakage	
27.	Check the tightness of all bolts and thread locking compound to be provided in bearing base bolts.	Provided	Provided	Provided	
28.	Examine the roofline conductors for defects/looseness at joints.	No defect	No Defect	No Defect	
29.	Check the roof for any foreign material such as wire / clothpieces.	No foreign material	OK	OK	
30.	Check tightness, cleanliness and wear-tear of shunt connection between Panto and Roof line.	Tight, clean	Tight clean	Tight Clean	7/10/42 06/08/23
31.	Check that the pan head locking pins springs are in place and secure. Check that the head moves freely on its mounting pins attached to the apex frame. If movement is not free, check that the head is not twisted, the side beam are not bent/twisted & mounting pins are not loose or bent.	OK	OK	OK	
32.	Check tightness of the flexible shunt connecting screws. When a loose screw is found the shunt should be disconnected from the head &/or the pantograph frame, the mating surface cleaned & the shunt reconnected with new screws. Check all shunts for frayed strands and renew any shunt in which 5% or more of the strands are frayed.	OK	OK	OK	
33.	AC Sch: Clean filter element; if found brownish then replace with new.	Clear	Clean	Clear	

Signature and Name of JE/SSE

SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI

IA/IB/IC Schedule of 3 Phase Electric Locomotive

लोको संख्या Loco No 32456 शिड्यूल में लेनेका दिनांक Date taken under Schedule 3.8.25

Sr. No	Details of Work / Inspection (Schedule)	Action Taken	Name of TCN	Sign / Remarks
A	VCB			
1	Check insulator for crack, VCB roof bus bar, chips and flash marks. If damage in service, renew it	Check		
2	Clean insulator, other parts that are dirty with soft, clean & dry cloth (Do not use cleaning products based on fluoridated or chlorate compounds or sodium meta silicate), apply Silicon Spray.	Clean		
3	Check contact spring (Earthing contact)	Check		
4	Check moving contact lever of auxiliary switch and tension spring & its mounting bolts for breakage	Check	Diomke 03.8.25	
5	Check the cables & lugs for broken cables, lugs looseness	Check		
6	Check the main bushing condition, its connection and foundation bolts and clamps.	Check		
7	Check the tightness of HV Connections (Torque 70Nm)	Check		
8	IC Sch: a) Check for damage to connection of the earthing isolator, Cleaning & greasing, if necessary. b) Check the torque of VCB fixing screw (Torque 70Nm) c) Remove the bottom cover and check that the control air and electrical connection are tightened & Ensure that there is no leakage from pressure regulator	Check Check ok		
B	MAIN TRANSFORMER			
1	Read off the oil level on the gauge situated on the conservator. Top up the oil as necessary and check for any signs of leakage. Clean the gauge with dry cloth	checked		
2	Inspect the colour of the silica gel. If it is pink, replace with blue silica gel	pink — changed		
3	Examine the flanges of the pipe couplings and flexible hose that link the transformer and conservator	checked		
4	Check the availability of hosepipe over the oil pipe compensator and check stoochi coupling pipes for proper layout/fitment.	checked		
5	Check the main Transformer and its protection cover for any damage, crack & oil leakage. Also refer instructions contained in RDSO's letter **no. EL/3.2.1/3-Ph / TC-0076** for arresting oil leakages cases"	checked		
6	Check the deformity of Transformer drain cock protection cover guard if deformed replace it	checked		
7	Perform the sample test on transformer oil. Check specific value of BDV, DGA, moisture and acidity and condition monitoring of loco transformer by Dissolve Gas Analysis. **(RDSO/SMI/138 & OEM Doc. Dt. 27th Nov, 1995)**	checked		manty 3.8.25
8	Check visually the condition of transformer foundation bolt and Nylon lock nuts for proper locking.	checked		
9	Check the Ecable. Clean and visually inspect the insulator condition and Electrical connection of transformer and converter for crack & flashed mark and ensure red marking	checked		
10	Visually inspect for leakage, damage, burning/overheat signs and all fixing clamps condition of oil cooling metallic pipes, Prismatic level gauge, TFP bushing, HV bushing, Hose pipe etc.	checked		

Signature and Name of JE/SSE

Sr. No.	Details of Work / Inspection (Schedule)				Action Taken	Name of TCN	Sign / Remark
C	EARTHING CONNECTION						
	a) Check [illegible]				Checked ok		
	b) Connection [illegible] type WAP5 [illegible] as per RDSO 2007/EL/SV [illegible] Rev.'0' dated 22.11.2007				Checked ok		Hemanth 9, 8 [illegible]
	c) Check [illegible]						
	Axle box 1	ok	Axle box 4	ok			
	Axle box 5	ok	Axle box 8	ok			
	d) Check Earthing [illegible] Bogie						
	Axle box 2	ok	Axle box 3	ok			
	Axle box 6	ok	Axle box 7	ok			

Signature and Name of JE/SSE

LOCO NO. 32456/JC

Sr.	Defects	Action Taken	Attended By
1	Cab1 fire Exiguisher Leakage from cock. —	attend	Jmithes
2.	Cab2 Both Hose Pipe Expired	attend	
3.	Cab1 FP Hose Pipe Expired.	attend	
4.	Both Silica gel is Pink. — changed —		
5.	CP2 Pressure Build up close at 8.5kg/cm2	attend	Jeevan
6.	CP2 Safety valve NOT Blowing		
7.	AR Tank Leakage from gaskit.	Attend	Jeevan
8.	Cab1 LP side wiper mechanical handle uncouple.	attend	
9.	Cab1 ALP side wiper arm Alnky Bolt loose.		
10.	Cab1 wiper water spray NOT wkg.		
11.	Cab2. LP side wiper water spray Leakege from wiper foundation & broken colon (d/repair)		
12.	Cab1 BP/FP & Safety guard bend condition.	attend	
13.	Both side Co-action and Emergency NOT wkg.		
14.	all Sander NOT Wkg.		
15.	Cab2 side MR & AC and cab-1 side BC EQ pipe not provided.		
16.	Panto 2 to check as d/repair — not ~~stop~~ raising.		
17.	Oil spread below SR-01 in Machine room.		
18.	All sand pipe nozzle missing or damaged.		
19.	W/N 5, and 8 non sand pipe missing (sandbox)		
20.	W/N 1, 2, 8 sandbox dummy missing.		
21.	W/N 2, sandbox air pipe uncoupled.		
22	W/N 6, 7 sandbox air pipe missing; all other available are flexible pipe		
23.	W/N 1, 2, 6, 7, 8, 11 sandbox safety bracket missing.		
24.	W/N 7, 8 sandbox cover 1 bolt missing.		
25.	CP1 drain plug (IC) — safety bracket missing.		